CHAPTER 12

The Cognitive Revolution

THE "COGNITIVE REVOLUTION" IN PSYCHOLOGY EMERGED FROM postwar developments in information theory and computer science. The development of the electronic computer created a new and technically proven model of the mind as a mechanical information processor, conceived of as operating on the same sorts of "rules and representations" employed by "intelligent" machines (Bechtel, 1988).

One of the peculiarities of the cognitive revolution was that many of its pioneers came to conceive of their own intellectual achievement in terms of Thomas Kuhn's (1970) analysis of the structure of scientific revolutions, according to which one general theoretical or methodological paradigm is replaced by a radically different paradigm, under the pressure of accumulating empirical anomalies (Lachman, Lachman, & Butterfield, 1979). Kuhn's influential *The Structure of Scientific Revolutions* was first published in 1962, as the new forms of cognitive psychology were being advanced and developed by Jerome Bruner (1915–), George Miller (1920–), Ulric Neisser (1928–), Allen Newell (1927–1992), and Herbert Simon (1915–2001). As James J. Jenkins (1923–) later remarked, during the early years of the cognitive revolution in psychology, "everyone toted around their little copy of Kuhn's *The Structure of Scientific Revolutions*" (Jenkins, quoted in Baars, 1986, p. 249).

Although there was no revolution in the strict Kuhnian sense, the development of cognitive theories from the 1950s and 1960s onward did mark a genuine discontinuity with behaviorist theories, including later "liberalized" neobehaviorist theories in terms of internal "mediating" r-s sequences (N. Miller, 1959; Osgood, 1957). Although the primary stimulus for the cognitive revolution came from without, the empirical problems faced by neobehaviorism in the 1950s and 1960s created an intellectual climate that left many psychologists predisposed to theoretical and methodical change. As Jenkins put it, "things were boiling over . . . a new day was coming" (Jenkins, quoted in Baars, 1986, p. 249).

INFORMATION THEORY

The primary stimulus for the growth of cognitive psychology came from outside academic psychology, notably from developments in logic, mathematics, and computer science, which were themselves a product of applied research on radar, message encoding, and missile guidance conducted during the Second World War.

Claude Shannon: Communication Theory

The wartime need to transmit maximum coded information in limited capacity channels promoted the development of information theory. Claude E. Shannon (1916–2001), an MIT engineering graduate who worked for Bell Laboratories, developed a mathematical theory of communication based upon the transmission of information from a "source" through a "channel" to a "destination." His goal was to identify the most efficient means of transmitting information via media such as telephone circuits and radio waves. Shannon measured information in terms of reduction in uncertainty and treated the **bit** (short for "binary unit") as the elemental unit of information: the amount of information required to determine between two equiprobable alternatives (Shannon, 1948). Shannon and Warren Weaver (1894–1978) developed statistical theories that described the relationships between variables in a communication system and the process of information flow though such a system (Shannon & Weaver, 1949). They were particularly concerned with the problem of loss of information as a signal is transformed in the course of transmission from source to destination, when it has to compete with background **noise**, defined as any random disturbance superimposed upon a signal (such as electrical noise caused by heat in electrical circuits).

George Miller, one of the pioneers of the cognitive revolution, introduced the statistical measures of information theory to psychology (Miller, 1953; Miller & Frick, 1949) and employed them in his analysis of language in *Language and Communication* (1951). Although information theory was first employed in psychology in the analysis of stimulus-response learning (Miller & Frick, 1949), units of information soon threatened to displace the stimulus-response connection as the primary focus of scientific psychology.

Shannon's binary analysis of units of information was based upon an assumed analogy between the binary values of symbolic logic (true/false) and switching circuits (on/off), which he described in his MIT master's thesis "A Symbolic Analysis of Relay and Switching Circuits" (1938). This analogy became the electrical engineering foundation for the later development of digital computers, or "logic machines."

Norbert Wiener: Cybernetics

Norbert Wiener (1894–1964), who had been a student of Bertrand Russell's at Cambridge, also made important contributions to information theory. During the war, he worked on guidance systems for torpedoes, which employed feedback from sound detectors to adjust their trajectory. He described the behavior of such mechanical but purposive devices in *Cybernetics* (1948). He defined **cybernetics** (derived from the Greek term for "steerman") as the scientific study of control and communication in animals and machines, which he analyzed in terms of the acquisition, use, retention, and transmission of information. Wiener characterized purposive behavior as the intelligent adjustment of behavior to environmental change, echoing earlier functionalist accounts (Angell, 1907), and complained that a major obstacle to the development of mechanical systems that could "mimic" purposive behavior in humans was the inadequacy of neobehaviorist theory.

In contrast to Hull, who tried to provide mechanistic explanations of the apparently purposive behavior of humans and animals, Wiener, in a paper that he co-authored with Arturo Rosenblueth and Julian Bigelow in 1943, maintained that the behavior of **servomechanisms** such as torpedoes is intrinsically purposeful:

> Some machines . . . are intrinsically purposeful. A torpedo with a target-seeking mechanism is an example. The term servomechanisms has been coined precisely to designate machines with intrinsic purposive behavior.
>
> —(Rosenblueth, Wiener, & Bigelow, 1943, p. 19)

He claimed that such machine behavior requires a teleological explanation in terms of regulation by **feedback**, defined as "signals from the goal that modify the activity of the object in the course of the behavior'" (Rosenblueth, Wiener, & Bigelow, 1943, pp. 19–20), and that this form of explanation applies to both living and mechanical systems:

> The behavior of some machines and some reactions of living organisms involve a continuous feed-back from the goal that modifies and guides the behaving object.
>
> —(Rosenblueth, Wiener, & Bigelow, 1943, p. 20)

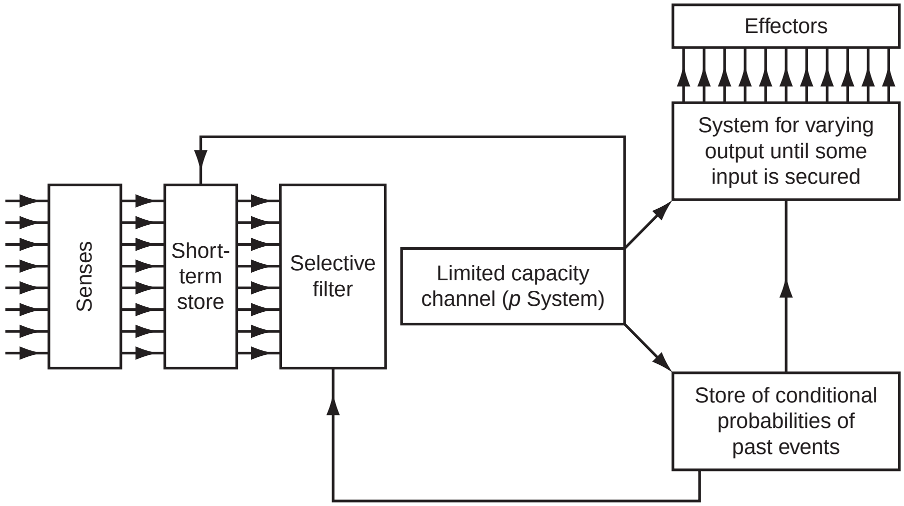
Wiener maintained that the same information-theoretical principles of explanation apply to the restricted class of animal, human, and machine behavior that involves regulation by information feedback. In doing so, he made a major contribution to the development of the cognitive revolution. He legitimized the concepts of purposive behavior and teleological explanation for hard-nosed scientific skeptics by demonstrating their applicability to the behavior of inanimate machines, the paradigm of mechanistic explanation since Descartes. Wiener also championed the autonomy of cognitive psychological explanation with respect to neurophysiology and physiology, by arguing that the same principles of teleological explanation apply to the purposive behavior of different material systems, whether composed of cells or silicone. As he put it, “Information is information, not matter or energy” (1948, p. 132). In this respect Wiener returned to the Aristotelian functionalist conception of the relation between psychological capacities and their modes of material instantiation.

In documenting the various ways in which animals, humans, and machines can employ information feedback, Wiener introduced many of the concepts that were later to play a significant role in computer science and cognitive psychology, such as “working memory” and “executive function.” He maintained that the ability of any mechanical system to engage in purposive behavior was based upon structural features suitable for the “acquisition, use, retention, and transmission of information” (1948, p. 161).

Between 1946 and 1953, the Macy Foundation funded twice-yearly conferences on cybernetics in New York, which were attended by neurophysiologists, logicians, statisticians, engineers, biologists, anthropologists, and social psychologists. Although interest in cybernetics declined in the 1950s, the meetings demonstrated the interdisciplinary appeal of cognitive theories based upon the flow of information, presaging the later interdisciplinary enterprise that became cognitive science.

Donald Broadbent: Information Processing

Psychologists originally employed information theory in the statistical description of information flow in communication systems, which reached a high-water mark with Wendell Garner’s *Uncertainty and Structure as Psychological Concepts* in 1962. However, they soon developed theories of information processing to explain cognitive processes. One of the first to do so was the British psychologist Donald Broadbent (1926–1993), who used the language of information processing in his theory of selective attention for auditory messages. In “A Mechanical Model for Human Attention and Immediate Memory” (1957), Broadbent treated human short-term memory as a “limited capacity channel” and provided a theoretical account of attention in terms of the active processing of information rather than the passive reaction to stimuli. He developed this account in *Perception and Communication* (1958), in which he recommended that the behaviorist language of stimulus and response be replaced by the language of information theory. For Broadbent, this was not a mere linguistic convenience, since he held that “the performance of selective listeners seems to vary with information as defined by communication theory, rather than with amount of stimulation in the conventional sense” (1958, p. 15). He maintained that theories of attention needed to “distinguish between the arrival of a stimulus at the sense-organ and the use of the information it conveys” (1958, p. 59).



Information-processing system.

Broadbent, D. E. (1958). Perception and communication. Oxford: Pergamon Press.

Broadbent stressed that one of the virtues of the language of information processing was that it preserved the autonomy of cognitive psychological explanation by enabling theorists to describe cognitive psychological processes without committing themselves to the details of the material systems in which they were instantiated (be they humans, animals, or machines). Broadbent claimed that one of the weaknesses of Hebb's (1949) theory of pattern recognition was that it was tied to a very specific neurophysiological hypothesis, which made his theory hostage to falsification by recalcitrant neurophysiological data. Yet, as Broadbent pointed out, Hebb's theory might "well be true even though the elements are not physically what Hebb supposed them to be" (1958, p. 306). He noted that this was precisely the fate of Gestalt psychology, which was rejected by most psychologists because of its link to an untenable theory of neural fields: "Their unlikely physiology has produced neglect of their genuine psychological achievements" (1958, p. 306). Broadbent's own cognitive psychological theories were neutral with respect to neurophysiology and were designed to provide cognitive psychological explanations of "what happened inside a man which was not a mentalistic introspective language, which was not hypothetical neurophysiology, and which wasn't simply a description of the visible behavior" (in Cohen, 1977, p. 63). Like Wiener, Broadbent championed the Aristotelian functionalist conception of cognitive capacities and their modes of material instantiation.

Broadbent's own theoretical perspective had developed from wartime research on human vigilance, which focused on the performance of human operators of radar, calculating machines, and aircraft. Wartime concerns with the complex behavioral repertoires demanded of pilots and machine-gunners forced psychologists to recognize the inadequacy of neobehaviorist accounts of skilled performance in terms of habit hierarchies, which had been favored in prewar industrial psychology. They developed theories of skilled performance in terms of the flexible adjustment of behavior governed by central control structures (Bartlett, 1943; Craik, 1947), akin to those developed by Lashley and Wiener in their accounts of serially ordered and intrinsically purposive behavior. As Broadbent put it in *Perception and Communication,*

> The picture of skilled performance built up by modern researches is one of a complex interaction between man and environment. Continuously the skilled man must select the correct cues from the environment, take decisions upon them which may possibly involve prediction of the future, and initiate sequences of responses whose progress is controlled by feedback, either through the original decision-making mechanism, or through lower-order loops. The processes of filtering the information from the senses, of passing it through a limited capacity channel, and of storing it temporarily are only part of the total skilled performance.
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> —(1958, pp. 295–296)

In the United States E. B. Hunt extended and developed Carl Hovland's information-theoretic analysis of concept learning (Hovland, 1952) into a full-blown theory of information processing in *Concept Learning: An Information Processing Problem* (Hunt, 1962), in which he treated concept learning as an active decision process operating on cognitive hypotheses. George Miller published a classic paper on the limited capacity of attention and memory as information channels titled "The Magical Number Seven, Plus or Minus Two: Some Limits on Our Capacity for Processing Information" (Miller, 1956). He demonstrated the limited capacity of sensory judgment, tachiscopic perception, and short-term memory and reprised in information-theoretical terms Wundt's earlier claim that the capacity of human attention is restricted to around seven units (Wundt, 1912).

Miller also demonstrated the limitations of the pure information-theoretical approach. Like Wundt, he noted that constraints on memory capacity can be surmounted by recoding information into meaningful "chunks" such as "mother" rather than the units "r" "h" "m" "t" "e" "o" (which can be rearranged to form the meaningful word "*mother*"):

> We must recognize the importance of grouping or organizing the input sequence into units or chunks. Since the memory span is a fixed number of chunks, we can increase the number of bits of information that it contains simply by building larger and larger chunks, each chunk containing more information than before.
>
> In the jargon of communication theory, this process would be called *recoding*.
>
> —(1956, p. 93)

Thus Miller, who had originally introduced information theory to psychology (Miller, 1951, 1953; Miller & Frick, 1949), was also instrumental in developing information-processing theory. He "set the agenda for the next phase of cognitive psychology in which information-processing concepts went beyond the confines of information theory" (Baddeley, 1994, p. 353).

Computers and Cognition

The most influential form of information-processing theory was the product of the creation of the electronic computer. The use of mechanical devices to perform mathematical operations goes back to the ancient Babylonians, and the idea of employing machines to process symbols was developed by Leibniz and Charles Babbage (1791–1871) in the 18th and 19th centuries. In the late 19th and early 20th centuries, George Boole (1815–1864), Gottlob Frege (1848–1925), and Bertrand Russell (1872–1970) developed the **new logic**, which considerably advanced the theoretical potential of symbol-processing machines. The significance of this logic was that it was truth-functional: Sentential operators such as "if . . . then" were defined in terms of primitive operators such as "and" and "not,"[1] which could be physically instantiated as on/off stitches in electrical circuits. Claude Shannon was the first to recognize the relation between the binary systems of symbolic logic and electronic circuitry and thus to explore the possibility of a "logic machine" (Shannon, 1938). McCulloch and Pitts (1943) developed a similar binary analysis of neural networks, whose operations they represented by the hypothetical firing or nonfiring of individual neurons.

Turing Machines In his attempt to answer the question of whether there could be a decision procedure for determining whether mathematical propositions are provable, the English mathematician Alan Turing (1912–1954) developed the abstract idea of a hypothetical machine, now known as a **Turing machine**, capable of performing elementary operations on symbols printed on a paper tape (Turing, 1936). He employed this notion to determine the computable numbers, defined as those printable by a Turing machine. Turing originally approached the question by conceiving of a human operator, or human "computer," who mechanically performed a "set of instructions" from a rulebook. By imagining a human replaced by a machine with a stored "set of instructions," Turing created the notion of a computer **program**.

Turing machines are defined by their "tables of instructions" or "programs," which specify their operations. Turing also conceived of a general-purpose machine, or **universal Turing machine**, capable of performing operations specified by the

[1]Thus "if p then q" could be defined as "not" (p and not q).

programs of a number of different Turing machines. In conceiving of a machine capable of performing a variety of tasks (such as computing numbers, drawing logical implications, and playing chess), Turing developed the principle of the modern computer (a decade before fast electronic calculating machines were produced).

Turing also explored an idea that was later to inspire American pioneers of cognitive psychology and cognitive science: that computing machines could be programmed to simulate human cognitive processes. One implication of Turing's theoretical analysis of computation was that any set of procedures specifiable in a binary code could be instantiated in a mechanical computer. Since Turing believed that the logical operations involved in human cognitive processing could be specified as a series of steps in binary code, he believed that a mechanical computer could simulate them. He had originally conceived of a Turing machine as a machine designed to perform tasks that could be performed by a human following a rulebook of instructions. As he later put it, "The idea behind digital computers may be explained by saying that these machines are intended to carry out any operations which could be done by a human computer" (1950, p. 436).

During the Second World War, Turing served with British Intelligence at Bletchley Park, where he and his colleagues used calculating machines to decipher German cipher codes, including the Enigma Code used by U-boats. At the end of the war he submitted his design for an "Automatic Computing Engine" (or for "building a brain," as he put it to a colleague) to the National Science Laboratory (Turing, 1946/1992). Although his ambitious scheme was formally accepted, it was never implemented (Hodges, 1992).

At the outbreak of the war, Turing converted his savings into silver bullion bars and buried them in the countryside surrounding Bletchley Park, but forgot where they were buried when he returned to retrieve them at the end of the war. A tormented homosexual trapped in a society in which homosexuality was still a criminal offense, he was forced to undergo treatment of his "unnatural" tendencies with injections of estrogen. Turing committed suicide in 1954 at the age of 42 by eating a cyanide-poisoned apple (Hodges, 1992).

ENIAC and EDVAC During the war, high-speed calculating machines were employed not only to decipher enemy codes, but also to handle the complex mathematical computations required for the operation of anti-aircraft guns and navigation systems. In engineering terms, the modern electronic computer was born when such devices became self-regulating: when machines operating on stored sets of instructions (programs) replaced human operators (Hunt, 1994).

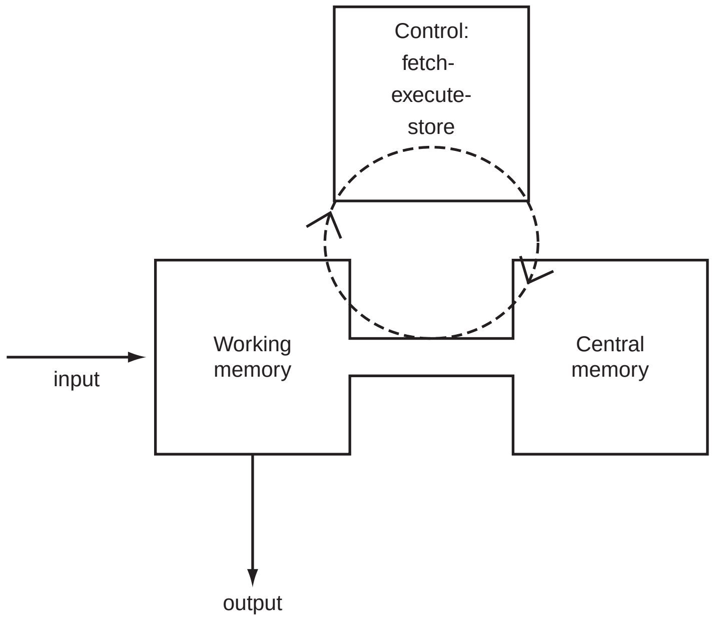
The first fast electronic calculating machine or computer, called ENIAC (Electronic Numerical Integrator and Automatic Computer), was constructed at the University of Pennsylvania in 1948 (it was originally designed to calculate artillery tables). John von Neumann (1903–1957), the Princeton mathematician and polymath, was a consultant on the project. He was responsible for a number of practical and conceptual innovations in the development of **digital computers**, which perform operations on binary units of information.[2] Von Neumann distinguished between the **software** and **hardware** of a computer: between the set of rules or instructions encoded by a computer program and the different physical systems that instantiate these rules or instructions (Dupuy, 2000). This distinction was a computational variant of Aristotle's functionalist conception of psychological capacities, since it allowed that one and the same program (software) could be multiply realized in a variety of different physical systems (hardware), including biological systems (sometimes known as **wetware**). For example, a variety of different physical machines (such as PCs and Macs) and humans can instantiate the same programs for adding numbers or playing chess.

Von Neumann was also one of the first to draw attention to the analogy between the organization of the brain and the circuitry of digital computers in his paper "The General and Logical Theory of Automata" (1951), first read at the Hixon Symposium in 1948. Although the original ENIAC machine performed operations in parallel as the human brain does, von Neumann later developed a machine that could perform operations serially, called EDVAC (Electronic Discrete Variable Automatic Computer). Von Neumann's design became the standard for computers in the following decades, with the consequence that modern computers, which employ a central control unit to read and execute programmed instructions serially, are often characterized as **von Neumann machines**. These machines were originally employed to perform mathematical operations, but the introduction of information-processing languages (IPLs) stimulated the development of computer programs for problem solving, decision making, and game playing.

Computer Simulation of Cognitive Processes In 1954, Allen Newell, who worked at the Systems Research Laboratory of the RAND Corporation, began work on the design of a program that would enable a computer to play chess. J. C. Shaw and Herbert A. Simon joined him in 1955. They worked together on the design of programs for JOHNNIAC (John von Neumann's Integrator and Automatic Computer), a computer built for the RAND Corporation in the 1950s. They temporarily abandoned the plan to develop a chess-playing program and instead directed their attention to simpler programs for proving theorems in geometry and logic.

Newell and Simon developed a program known as the **Logic Theorist**, based upon a specially developed information-processing language. The first machine proof of a theorem in symbolic logic, Theorem 2.01 from Russell and Whitehead's *Principia Mathematica*, was produced on the JOHNNIAC computer in August 1956.

[2]In contrast to analog computers, which represent variable physical processes (such as changes in temperature) via changes in some physical variable (such as electrical potential).



Generic computer architecture (von Neumann machine).
Harnish, R. M. (2002). Minds, brains, computers. Oxford: Basil Blackwell

Newell and Simon first publicized their achievement at the Symposium on Information Theory held at MIT in September 1956, the conference at which the cognitive revolution was born, according to Bruner (1980) and Miller (1989). The following year, Newell, Shaw, and Simon began work on a more complex program, the **General Problem Solver** (GPS), capable of performing a wider range of cognitive tasks, such as playing chess and problem solving, in addition to proving theorems in geometry and logic.

Newell, Shaw, and Simon were convinced of the relevance of their research to the cognitive psychology of human problem solving, and they saw their machine proofs as the simulation of human information processing. They presented their case to psychologists in "Elements of a Theory of Problem Solving," published in *Psychological Review* in 1958. Newell, Shaw, and Simon offered a theory of the "control-systems" of humans and machines to explain their "problem-solving behavior" in terms of "information processing." They explained human and machine problem-solving behavior in terms of the information processes responsible for such behavior: "An explanation of an observed behavior of the organism is provided by a program of primitive information processes that generates this behavior" (1958, p. 151).

Newell, Shaw, and Simon were careful to stress that their theory of information processing in terms of rule-governed operations on symbols had nothing essentially to do with electronic digital computers and that programs specifying such operations could be developed independently of digital computers. They noted that such programs had been developed earlier by Adriaan de Groot (1914–2006), in his classic analysis of problem solving in chess (de Groot, 1946), and by Otto Selz (1922), the last major theorist of the Würzburg school (who was de Groot's teacher):

> Our position is that the appropriate way to describe a piece of problem solving behavior is in terms of a program: a specification of what the organism will do under varying environmental circumstances in terms of certain elementary information processes it is capable of performing. This assertion has nothing to do—directly—with computers. Such programs could be written (now that we have discovered how to do it) if computers had never existed.
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> —(Newell, Shaw, & Simon, 1958, p. 153)

The Logic Theorist employed a program that comprised four rules of inference (substitution, replacement, detachment, and chaining), which enabled it to employ a variety of methods of "discovering" proofs. With the program and the axioms of *Principia Mathematica* stored in memory, the Logic Theorist was presented with the first 52 theorems of Chapter 2 of *Principia Mathematica*. It succeeded in proving 38 of them.

Like Wiener, Newell, Shaw, and Simon insisted that their theory applied indifferently to humans and machines: "These programs describe both human and machine problem solving at the level of information processes" (1958, p. 153). They claimed that their theory of information-processing was autonomous with respect to human neurophysiology and maintained a functionalist conception of the relation between programs of information-processing and the human brains and machines capable of realizing them:

> Problem solving—at the information-processing level at which we have described it—has nothing specifically "neural" about it, but can be performed by a wide class of mechanisms, including both human brains and digital computers. We do not believe that this functional equivalence between brains and computers implies any structural equivalence at a more minute anatomical level (e.g., equivalence of neurons with circuits). Discovering what neural mechanisms realize these information-processing functions in the human brain is a task for another level of theory construction. Our theory is a theory of the information processes involved in problem solving, and not a theory of neural or electronic mechanisms for information processing.
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> —(1958, p. 163)

Although Newell, Shaw, and Simon claimed to have offered "a thoroughly operational theory of human problem solving" (1958, p. 166) and to have eliminated "the vaguenesses that have plagued the theory of higher mental processes" (1958, p. 166), they did not offer an operationally defined theory of problem solving. The "operational" in "operational theory" was a reference to the operations performed on symbols by programs, which were defined "in terms of elementary information processes," such as the rules governing the substitution, detachment, replacement, and chaining of logical symbols (1958, p. 157). In contrast to neobehaviorists, Newell, Shaw, and Simon did not believe that the question of "what is learned" in problem solving was a pseudo-issue (Kendler, 1952), but instead offered a specific and substantive answer to the question. They claimed that the Logic Theorist learned theorems, as well as subproblems for proofs that had been previously attempted, and which theorems were useful in conjunction with particular methods.

They maintained that the Logic Theorist simulated problem solving in humans by executing "the same sequences of information processes that humans execute when they are solving problems" (1958, p. 153). As evidence for this, they noted that the Logic Theorist not only succeeded in solving a range of problems solvable by humans (no mean achievement in itself), but that it also manifested many of the characteristics "exhibited by humans in dealing with the same problems" (1958, p. 162). These included dependence on the sequence in which problems were presented, preparatory and directional "set," employment of heuristically governed vicarious trial and error, and hierarchical organization of problems and subproblems (1958, p. 162).

Newell, Shaw, and Simon later recognized that the Logic Theorist did not mimic the behavior of humans particularly well when compared with subject protocols based upon "think-aloud" tapes of human subjects engaged in the solution of logical problems (devised by O. K. Moore and S. B Anderson in their research on human problem solving at Yale University). The program of the later General Problem Solver was specially designed to reproduce the distinctive features of human problem solving revealed through human subject protocols (Newell & Simon, 1972).

Artificial Intelligence The Logic Theorist heralded the development of "information-processing" or "computational" psychology and represented an early achievement of the emerging discipline that came to be known as **artificial intelligence**: the science of intelligent machines (Minsky, 1963). This naturally raised the question of whether cognitive processes, such as those involved in proving theorems, playing chess, or understanding language, can properly be ascribed to machines, in the same sense in which they are paradigmatically ascribed to humans. This question has vexed psychologists as much as it has vexed critical philosophers and humanists.

Turing, the intellectual pioneer of the modern computer, was himself optimistic about the prospect of developing genuinely intelligent computing machines.

In "Computing Machinery and Intelligence" (1950), he described an "imitation game," now known as the **Turing test**, in which a human interrogator poses questions to a machine and another human, whose only contact with them is via a teleprinter link. Turing maintained that we should be prepared to ascribe intelligence to a machine if we could not discriminate the responses of the human communicator from a machine simulating or "imitating" these responses.

The Chinese Room While many workers in artificial intelligence have accepted this "solution," psychologists and philosophers such as John Searle (1980) and Herbert Dreyfus (1972) have been much less enthusiastic. Searle has argued that the symbols that computers manipulate, such as those processed by computers operating on "story-reading" programs (Schank & Abelson, 1977), mean nothing to them, unlike the symbols that humans employ in language comprehension and communication.

To illustrate his point, Searle imagined a room in which a person who speaks only English receives written questions in Chinese. The person responds with printed answers in Chinese in accord with instructions from an English rulebook, which specifies the appropriate Chinese answers to questions set in Chinese. The person identifies the Chinese questions merely by the shape of the symbols and prints out the set of symbols that the rulebook identifies as the appropriate answer in Chinese. Searle claimed that a person performing such a task could pass the "Turing test," but would not understand Chinese: Such a person would merely "produce the answers by manipulating uninterpreted formal symbols " (Searle, 1980, p. 418). Since such a person would essentially "behave like a computer," performing "computational operations on formally specified elements," Searle concluded that linguistic comprehension and communication should not be ascribed to digital computers.

Searle's argument presents a real challenge to artificial intelligence researchers committed to what Searle calls **strong AI**: the view that we can ascribe cognitive states to computers in exactly the same sense that we ascribe them to humans.[3] Such researchers have been obliged to specify what more is required of mechanical systems for the symbols that they process to have meaning for such systems. Popular candidates are sensory and motor links to the physical world and/or some form of learning or analogue of learning.

However, Searle's challenge presents much less of a problem for cognitive psychologists. Despite the initial enthusiasm for computer simulation among psychologists (Hovland, 1960), Simon's confident prediction that most theories in psychology would eventually take the form of computer programs (Dreyfus, 1972)

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[3]Strong AI is contrasted with weak AI, the view that computers are useful tools for formulating and testing hypotheses about the mind or the view that we can create machines that can perform tasks such as calculation and industrial product control, which would be described as intelligent if performed by humans.

turned out to be false. Cognitive psychologists never really embraced computer simulation as a means of testing theories of human cognitive processing, even as they developed detailed theories of human cognitive processing inspired by computer models.

Indeed, it may be fairly said that computer simulation and the computer model of cognition proved to be an example of the proverbial ladder that could be thrown away once psychologists were prepared to accept hypothetical theoretical constructs relating to cognitive states and processes (whose meaningful contents were independent of the content of operational definitions). Nonetheless, it would be hard to overestimate the powerful role played by the computer simulation of human cognitive processes in establishing the scientific legitimacy of causal explanatory appeals to cognitive rules and representations. The ability to specify cognitive theories as computer programs and to test them via the mechanical performance of computers gave the lie to behaviorist complaints about the vagueness and untestability of cognitive theoretical constructs. As Margaret Boden put it, the computer model promoted the double discipline of explicitness and testing: "a computer model enables us not only to state and clarify our theoretical ideas, but to find out whether they have the inferential consequences we believe them to have" (Boden, 1997, p. 56).

The demonstration that information-processing programs simulating human cognition could be instantiated in electronic machines completely undermined standard behaviorist complaints about theoretical appeals to cognition as a return to immaterial souls or spirits. Descartes' discredited ontological dualism was replaced by the respectable methodological dualism of von Neumann's distinction between cognitive software and mechanical hardware (or wetware, in the case of animals and humans). For cognitive psychologists, this functionalist conception of cognition enabled them to preserve the autonomy of cognitive psychological explanation without abandoning their common commitment to materialism. While they emphasized that theories about cognitive processes, like theories about computer programs, were not about material systems per se, they granted that they could be instantiated only in material systems of sufficient complexity (such as human and animal brains or computer hardware).

COGNITIVE PSYCHOLOGY

Other forms of cognitive theory were developed within psychology during the 1950s, more or less independently of the work on computer simulation and artificial intelligence, notably by Jerome Bruner, George Miller, and Ulric Neisser, who also played a significant role in the institutional development of cognitive psychology.

Jerome Bruner: Higher Mental Processes

Jerome Bruner, a student of William McDougall's at Duke and Gordon Allport's at Harvard, worked on propaganda, morale, and public opinion research during the Second World War. He explored the influence of cognitive factors such as expectation, emotion, and motivation on perception in a series of experimental studies (Bruner & Postman, 1947a, 1947b; 1949) that came to represent the "new look" in perception (Bruner & Krech, 1950). These studies indicated that subjects who are hungry or poor, for example, tend to see food or money when presented with ambiguous perceptual data. They included studies of perceptual defense (inspired by Freudian notions of repression) that indicated that taboo words are not as readily recognizable in tachiscopic presentations as neutral words, even though they generate measurable affective responses.

Bruner treated perception as an active cognitive process rather than the passive association of sensory stimuli. He developed this emphasis on the active role of cognition in *Studies in Thinking* (Bruner, Goodnow, & Austin, 1956), the first product of the collaborative Cognition Project that Bruner set up at Harvard in 1952. *Studies in Thinking* was an experimentally based study of the cognitive strategies that subjects employ in concept formation (such as "successive scanning," "conservative focusing," and "focus gambling"). This influential work deserves to be classified as the first substantive psychological monograph of the cognitive revolution.[4]

Bruner was explicit about his intentional break with the peripheralist stimulus-response behaviorist tradition and his return to a cognitive orientation or, as he put it, his "revival" of the study of "Higher Mental Processes":

> One need not look far for the origins of the revival. Partly, it has resulted from a recognition of the complex processes that mediate between the classical "stimuli" and "responses" out of which stimulus-response learning theories hoped to fashion a psychology that would by-pass anything smacking of the "mental." The impeccable peripheralism of such theories could not last long. . . .
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> . . . Information theory is another source of the revival. Its short history in psychology recapitulates the fate of stimulus-response learning theory. The inputs and outputs of a communication system, it soon became apparent, could not be dealt with exclusively in terms of the nature of these inputs and outputs alone nor even in terms of such internal characteristics as channel capacity and noise. The coding and recoding of inputs—how incoming signals are sorted and organized—turns out to be the important secret of the black box that lies athwart the information channel.
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> —(Bruner et al., 1956, pp. vi–vii)

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[4]Even though it was predated by Rapaport's *The Organization and Pathology of Thought* (1951) and Vinacke's *The Psychology of Thinking* (1952), not to mention Moore's 1938 *Cognitive Psychology*.

As in the case of other pioneers of the cognitive revolution, the original stimulus for Bruner's work came from outside American psychology. Bruner was influenced by the early work of the British psychologist Sir Frederic C. Bartlett (1886–1969) on memory schemas (Bartlett, 1932) and by the work of the European psychologist Jean Piaget (1896–1980), who had been studying the cognitive development of children since the 1920s (Piaget, 1926, 1927, 1930). Bruner met Bartlett while a visitor at the University of Cambridge from 1955 to 1956, where they organized a conference on cognition in the summer of 1956 (funded by the Rockefeller Foundation), the same year he first visited Piaget in Geneva. Piaget's influence was reflected in *Studies in Cognitive Growth* (Bruner, Oliver, & Greenfield, 1966), a work that promoted a theoretical conception of cognitive development as active and creative in contrast to Piaget's own theory of biologically determined stages. Bruner's account of cognitive development partly reprised earlier criticisms of Piaget's theory developed by the Russian psychologist Lev Vygotsky (1896–1934). Although both Piaget and Vygotsky (1934/1986) advanced cognitive theories in the 1920s, they had little influence on the early development of cognitive psychology in America in the 1950s (Bruner being a notable exception). English translations of the works of both authors became readily available in America only in the 1960s.

Bruner moved to Oxford in 1972, where he continued to work on the cognitive development of children. He returned to the United States in 1981 to take up an appointment at The New School of Social Research and later moved to Rockefeller University.

George Miller: Cognitive Science

George Miller majored in English and speech at the University of Alabama, and gained his PhD in psychology from Harvard in 1946. He was one of the first psychologists to embrace the statistical formulations of information theory developed by Shannon (Miller, 1953), although his early work was very much in the behaviorist tradition. Miller used statistical descriptions of information flow in his analysis of operant conditioning (Frick & Miller, 1951; Miller & Frick, 1949), which he extended to the analysis of verbal communication in *Language and Communication* (1951), one of the earliest sources of Skinner's views on verbal behavior. However, during the 1950s Miller came to treat humans more as (active) processors of information than as (passive) channels of information. He was impressed by the computer simulation of cognitive processes and was strongly influenced by his personal acquaintance with cognitive pioneers such as Bruner and Chomsky. Miller claimed that he abandoned behaviorism as a direct result of meeting Chomsky at a summer seminar at Stanford University (Miller, 1989).

Miller promoted Chomsky's ideas on language in "Some Psychological Studies of Grammar" (1962), having earlier reprised the experimental study of attention pioneered by Wundt (Miller, 1956). By 1965 Miller had abandoned the behaviorist mantra altogether and had joined with Bruner in talking

> about hypothesis testing instead of discrimination learning, about the evaluation of hypotheses instead of the reinforcement of responses, about rules instead of habits, about productivity instead of generalization, about symbols instead of conditioned stimuli, about linguistic structure instead of chains of responses.
>
> —(Miller, 1965, p. 20)

In 1960 Miller and Bruner managed to persuade McGeorge Bundy, the Dean of Faculty at Harvard, to establish the Center for Cognitive Studies (funded by the Carnegie Corporation), which hosted a number of distinguished visiting faculty engaged in the study of language, memory, perception, concept formation, and cognitive development. Although the life of the center was short (it was discontinued after 10 years, due to intellectual fragmentation and departmental politics), it provided temporary institutional legitimization of the cognitive "paradigm" in psychology at a critical period of its development. Miller moved to Rockefeller University and then to Princeton University, where he established a new program and laboratory in **cognitive science**, the name for the newly evolved interdisciplinary matrix of cognitive psychology, artificial intelligence, and linguistics (and associated disciplines such as neurophysiology, logic, mathematics, and philosophy).

Strategies, Programs, and Plans

Although the various programs of theory and research in psycholinguistics, attention, computer simulation, concept development, and cognitive growth proceeded largely independently of each other (at least in the early stages), they were united on the conceptual level by their commitment to a realist construal of theories about cognitive states and processes and by the joint acknowledgment by early cognitive theorists of each other's work. Chomsky, Broadbent, Newell, Shaw, Simon, Bruner, and Miller were committed to a realist conception of theories about cognitive states and processes as autonomous hypothetical constructs and made no attempt to provide operational definitions of them. Bruner cited the work of Herbert Simon (1962) in *Studies in Cognitive Growth* (Bruner, Oliver, & Greenfield, 1966); Newell, Shaw, and Simon treated Bruner's theoretical term "strategy" as referencing essentially the same type of cognitive rule structure as their theoretical term "program" (1958, p. 153); and Chomsky cited Bruner and Newell, Shaw, and Simon (and Lashley) in his 1959 review of Skinner's *Verbal Behavior* (Chomsky, 1959, pp. 55–57).

Miller, Eugene Galanter, and Karl Pribram integrated emerging cognitive theories in *Plans and the Structure of Behavior* (1960). They proposed a general framework for the development of information-processing theory based upon the notion of a *plan*, defined as a "hierarchical process in the organism that can control the order in which a sequence of operations is to be formed" (1960, p. 16). The general orientation of the book was heavily influenced by recent work in computer simulation, and the authors avowed that the term *program* could be substituted for *plan* throughout the work. Rejecting peripheralist accounts of behavior based upon the chaining of stimulus-response sequences, they recommended the TOTE (Test-Operate-Test-Exit) routine as a superior unit of analysis.

Ulric Neisser: Cognitive Psychology

Ulric Neisser's *Cognitive Psychology* (1967), which integrated recent work on perception, attention, concept development, psycholinguistics, and computer simulation, effectively christened the emerging field of cognitive psychology. Neisser's interest in cognitive psychology was stimulated by his work with Miller at Harvard (where he gained his PhD in 1956) and Köhler at Swarthmore (where he gained his MA in 1952). He taught at Brandeis and Cornell University before moving to Emory University in 1983.

Neisser articulated the new "information-processing" approach of cognitive psychology, distinguishing it from alternative physiological, psychoanalytic, and behaviorist approaches. He characterized "cognitive psychology" as the study of "cognitive mechanisms" and defined cognition as "all the processes by which sensory input is transformed, reduced, elaborated, stored, recovered and used":

> Such terms as *sensation*, *perception, imagery, retention, recall, problem-solving*, and *thinking*, among many others, refer to hypothetical stages or aspects of cognition.
>
> —(1967, p. 4)

Neisser also championed a realist conception of cognitive hypothetical constructs:

> The basic reason for studying the cognitive processes has become as clear as the reason for studying anything else: because they are there. Our knowledge of the world *must* be somehow developed from the stimulus input. . . . Cognitive processes surely exist, so it can hardly be unscientific to study them.
>
> —(1967, p. 5)

Neisser's cognitive psychology developed information-processing psychology beyond the confines of information theory and computer simulation. Like Miller and Broadbent, Neisser recognized that information theory is inadequate as a characterization of cognitive processes, since it treats cognitive systems as "unselective." In contrast, Neisser maintained that human beings "are by no means neutral or passive towards the incoming information. . . . they select some parts for attention at the expense of others, recoding and reformulating them in complex ways" (1967, p. 7). He granted that "the task of the psychologist trying to understand human cognition is analogous to that of a man trying to discover how a computer has been programmed" (1967, p. 6), but his theoretical exploitation of computational concepts did not involve any blind commitment to computer simulation, which he critically dismissed as "simplistic."

The Cognitive Revolution

Neisser's book introduced the cognitive psychological "paradigm" and the "cognitive revolution" (although Neisser himself never used the terms). Yet, like the behaviorist "revolution," the cognitive revolution was not an overnight affair. It began in the 1950s, but progressed slowly throughout the 1960s. This was despite Donald Hebb's call for a "second American revolution" in psychology devoted to the "serious, persistent, and if necessary, daring exploration of the thought processes "(1960, p. 745) and his confident claim that computational cognitive theories had begun to displace behaviorist learning theories:

> It is becoming apparent from such work as that of Broadbent (1958) and of Miller, Galanter and Pribram (1960) that the computer analogy, which can readily include an autonomous central process as a factor in behavior, is a powerful contender for the center of the stage.
>
> —(Hebb, 1960, p. 740)

Stevens's 1951 *Handbook of Experimental Psychology* and Woodworth and Schlosberg's 1954 *Experimental Psychology* had little to say about cognitive processes (Woodworth and Schlosberg's 1954 text had less on cognition than *Woodworth's* 1938 text), and the *Annual Review of Psychology* reported little on cognition throughout the 1950s and early 1960s (Hearnshaw, 1989).

Behaviorism remained institutionally as well as intellectually dominant in the immediate postwar period, when it was very hard to attain an academic position in psychology without at least an avowed commitment to behaviorism. As Miller, who began his career as a behaviorist, recalled,

> The chairmen of all the important departments would tell you they were behaviorists. Membership of the elite Society of Experimental Psychology was limited to people of a behavioristic persuasion. . . . The power, the honors, the authority, the textbooks, the money, everything in psychology was owned by the behaviorists. . . . those of us who wanted to be scientific psychologists couldn't really oppose it. You just wouldn't get a job.
>
> —(Baars, 1986, p. 203)

Yet by the late 1960s cognitive psychology had established itself as a viable theoretical and research program, and it was progressively institutionalized throughout the 1970s. New journals devoted to cognition were founded, such as *Cognitive Psychology* (1970), *Cognition* (1972), *Memory and Cognition* (1973), *Cognitive Science* (1977), and *Cognitive Therapy and Research* (1977). Departments began to hire faculty with specialization in cognitive psychology, undergraduate textbooks began to appear to satisfy the demand generated by new courses in cognitive psychology, and graduate programs in cognitive psychology (and interdisciplinary cognitive science) were introduced at a number of universities. By the late 1970s the cognitive revolution appeared complete, with Lachman, Lachman, and Butterfield (1979) declaring that a Kuhnian "paradigm shift" had occurred in psychology, in which the "information-processing" paradigm had displaced the earlier behaviorist paradigm:

> Information-processing psychology . . . has become the dominant paradigm in the study of adult cognitive processes.
>
> —(1979, p. 6)

They claimed that from the perspective of the "information-processing" paradigm, cognitive psychology was the study of "the way man collects, stores, modifies, and interprets environmental information or information already stored internally" (Lachman, Lachman, & Butterfield, 1979, p. 7).

By the 1980s the cognitive "paradigm" began to penetrate subdisciplines such as social, developmental, and clinical psychology. Social psychologists turned their attention to "social cognition" (Fiske & Taylor, 1982; Nisbett & Ross, 1980), developmental psychologists became engaged in debates over the nature and development of the child's "theory of mind" (Astington, Harris, & Olston, 1988; Perner, Leekam, & Wimmer, 1987; Wellman & Esters, 1986), and "cognitive therapy" became the new fashion in clinical psychology (Beck, 1976; Ellis, 1984).

Whatever their differences, and whatever their degree of commitment to the computer analogy, most cognitive psychologists from the 1970s onward were engaged in the study of cognitive processing. They exemplified the "rules and representations" (Bechtel, 1988) approach to cognition pioneered by Chomsky and given mechanical expression in the computer simulation of cognitive processes by Newell, Shaw, and Simon. This approach was reflected in a variety of seminal works, such as Anderson's *The Architecture of Cognition* (1983), Johnston Laird's *Mental Models: Towards a Cognitive Science of Language, Inference and Consciousness* (1983), Marr's *Vision: A Computational Investigation Into the Human Representation and Processing of Visual Information* (1982), Kosslyn's *Image and Mind* (1980), Schank and Abelson's *Scripts, Plans, Goals and Understanding* (1977), and Rosch and Lloyd's *Cognition and Categorization* (1978).

THE COGNITIVE REVOLUTION

Some have represented the development of cognitive psychology as a genuine revolution in psychological science (Baars, 1986; Gardner, 1985; Lachman, Lachman, & Butterfield, 1979). Others have questioned whether there was a revolution and suggested that cognitive psychology evolved naturally from "liberalized" forms of neobehaviorism that allowed for the introduction of internal intervening variables such as "mediating" r-s connections (Amsel, 1989; Holdstock, 1994; Kendler & Kendler, 1975; Leahey, 1992; Mandler, 2002).

The Cognitive Revolution as "Paradigm Shift"

Some have represented the emergence of cognitive psychology as a Kuhnian "paradigm shift" (Kuhn, 1970), in which the theoretical and methodological schema of behaviorism gave way to the theoretical and methodological schema of cognitivism under the pressure of irresolvable empirical anomalies (Lachman, Lachman, & Butterfield, 1979; Palermo, 1971; Weimer & Palermo, 1973). According to Lachman, Lachman, and Butterfield (1979), the development of information-processing theory in cognitive psychology represented a Kuhnian "scientific revolution," in which the "puzzle-solving" tradition of behaviorism was displaced by the "puzzle-solving" tradition of cognitive psychology:

> The scientific study of cognitive psychology has moved dramatically forward under the pre-theoretical commitments of the information-processing approach. Great progress is reflected in the way in which our discipline has refocused its research effort toward accounting for intelligent human behavior. This refocusing required a revolution. The significant issues simply could not be addressed adequately within the framework of neobehavioristic psychology.
>
> Now information-processing psychology is an established paradigm, and it guides the vast bulk of psychological research in human cognition. Our revolution is complete, and the atmosphere is one of normal science.
>
> —(Lachman, Lachman & Butterfield, 1979, p. 525)

However, for the following reasons, the historical movement from behaviorism to cognitive psychology is not best characterized in terms of a Kuhnian "paradigm shift." The various empirical problems faced by behaviorism, such as the difficulty of explaining linguistic behavior in terms of conditioning theory and the recognition of biological limits on conditioning, did not result in the abandonment of behaviorism. Behaviorists continued to maintain their in-house journals, their own APA division, and a sizable professional membership. Indeed, Skinner's school of radical behaviorism expanded institutionally during the same period in the 1950s that cognitive psychology was being developed (Krantz, 1972). The primary stimulus for the emergence of cognitive psychology came from external developments in information theory, linguistics, and computer simulation, rather than from the internal empirical anomalies faced by behaviorism.

The conflict between behaviorism and cognitive psychology was not a conflict between *exclusive* theoretical paradigms, on a par with the conflict between the physical theories of Newton and Einstein in the early 20th century or between the wave and particle theories of light in the early 19th century. The critical evidence that supported Einstein's physical theory and the wave theory of light appeared to demonstrate the general inadequacy of Newton's physical theory and the particle theory of light and led to their complete rejection by most scientists. Yet few imagined that the empirical problems faced by behaviorism demonstrated the fundamental inadequacy of theories of classical and operant conditioning. The recognized empirical anomalies indicated only a general delimitation of the scope of explanations in terms of conditioning (albeit long overdue) and the extension of biological and cognitive explanations to the range of human and animal behavior for which conditioning theory proved to be inadequate.

It was only because behaviorists had overestimated the scope of conditioning theory and presumed that it was able to accommodate all forms of animal and human behavior, including complex forms of human behavior such as linguistic behavior, that it faced these empirical problems. Earlier animal and comparative psychologists had not been so intellectually imperialistic. Lloyd Morgan had recognized biological limits on learning and the inability of theories of associative learning to explain higher cognitive processes (Morgan, 1894/1977, 1896).

One would be hard put to discriminate a set of core theoretical and methodological schema constitutive of a behaviorist paradigm, other than a commitment to observable behavior as the subject matter of scientific psychology. Watson, Tolman, Hull, and Skinner differed quite radically in their theoretical explanations of human and animal behavior and their advocacy of inductive and hypothetic-deductive approaches to scientific method. It is probably even harder to discriminate a set of theoretical and methodological schema constitutive of a cognitive paradigm, other than a commitment to cognition as a legitimate subject matter of scientific psychology. There are at best only family resemblances between the various theories advanced by Chomsky, Broadbent, Bruner, Simon, Miller, and Neisser, for example.

The cognitive revolution did not represent a revolution in terms of attitudes to the existence of cognitive states and processes. It is commonly supposed that behaviorists denied, whereas cognitive psychologists affirmed, the existence of cognitive states and processes (Baars, 1986; Fodor, 1975). Yet Watson, Tolman, Hull, and Skinner never denied the existence of cognitive states and processes (although all but Tolman denied that cognitive states and processes play a role in the causal explanation of human and animal behavior), and many early cognitive psychologists avowed agnosticism with respect to the existence of cognitive states and processes (Anderson, 1980; Lachman, Lachman, & Butterfield, 1979; Nisbett & Ross, 1981).

From Intervening Variables to Cognitive Hypothetical Constructs

Although the cognitive revolution did not constitute a Kuhnian paradigm shift, it did mark a significant change in the conception of theories of cognitive states and processes: from their treatment as operationally defined *intervening variables* to their treatment as independently meaningful *hypothetical constructs*. Even the most "liberalized" forms of neobehaviorist theory never possessed independent meaning. Theoretical postulates such as "drive," "habit strength," "pure stimulus act," and all the intervening variables of "mediation theory" were provided with rigorous operational definitions, since neobehaviorists avowed that independent (or surplus) meaning has no place in a properly scientific psychology. In contrast, the theories advanced by cognitive psychologists from the 1950s onward did possess independent meaning. Cognitive psychologists provided substantive theoretical definitions of the sensory register, attention, long- and short-term memory, depth grammar, cognitive heuristics, visual perception, propositional and imagery coding, episodic and semantic memory, template-matching, procedural networks, inference, induction, and the like, but eschewed operational definitions (as opposed to specified operational measures) of these cognitive states and processes.

However, the difference between neobehaviorist and cognitive psychological theories was more than a difference between operational definition and independent theoretical meaning. Unlike later cognitive psychological theories, neobehaviorist theories lacked specifically cognitive content. Although neobehaviorists did employ terms like *cognition, concept*, and *representation*, they did not use them to reference the types of contentful representational states that cognitive psychologists maintain are employed in the rule-governed processing of information.

For example, Hull operationally defined concept formation in terms of stimulus discrimination and verbal association. Thus a person was said to grasp the meaning of the concept "dog" when he or she could discriminate dog stimuli and associate the verbal label "dog" with them (Hull, 1920, p. 5). Yet parrots could achieve this form of discriminative learning without any mastery of concepts or linguistic meaning (Goldstein & Scheerer, 1941). Hull's operational definition of a concept in terms of a conditioned (verbal) response was quite different, for example, from the cognitive theoretical definition of a concept advanced by Homer Reed (1946), in terms of a word or idea that stands for any one of a group of things.

Analogously, Osgood (1953) defined a "representational mediation process" as an internal stimulus that elicits a behavior (or "fractional portion" of a behavior) normally elicited by a stimulus, in the absence of that stimulus (or prior to its appearance). For example, he claimed that an "internal stimulus" or "external verbal stimulus" associated with a stimulus such as a hammer comes to represent hammers. Yet such an internal stimulus or verbal label "hammer" only represents hammers in the sense that a tone "represented" food for Pavlov's dogs: It was a conditioned causal sign for it. As Osgood put it, such postulated internal states "represent" objects in the fashion that dark clouds "represent" rain (they are causal indicators of rain):

> The buzzer elicits an "anxiety" reaction—part of the "fear" response originally made to the shock—and it is by virtue of this fact that it means or represents shock.
>
> —(Osgood, 1953, p. 695)

Yet representational states employed in thought, memory, judgment, inference, and the like are symbols of objects, not causal signs for them. The concept of buzzer and the word *buzzer,* for example, represent and mean buzzers, independently of whether or not they have come to be associated with shock via conditioning. Even when they *have* come to be associated with shock via conditioning, the concept of buzzer and word *buzzer* continue to represent buzzers and not shock.

That is, there was nothing cognitive about the internal mediational states (or internal r-s sequences) postulated by neobehaviorists (with the exception of Tolman's cognitive maps). Hull and his followers seem to have supposed that cognitive states are merely internal states capable of generating behavior, which can be instantiated in mechanical devices or "psychic machines" (Hull, 1937). Yet this is plainly not the case. Although programmed computers may be able to simulate cognitive states and processes and generate "artificially intelligent" behavior, traditional mechanical devices, including the hydraulic statues in the Royal Gardens at St. Germain that so impressed Descartes, plainly cannot. The electromechanical models of classical and instrumental conditioning developed in the 1950s (e.g., Walter, 1953), in contrast to the problem-solving programs developed by Newell, Shaw, and Simon (1958), were not models of cognition. Hull seems to have supposed that the only alternative to his account of cognition in terms of conditioned learning was an appeal to "non-physical" entities such as immaterial souls or spirits (Hull, 1937, pp. 31–32), but the cognitive revolution itself demonstrated that it was not.

The development of independently meaningful cognitive theories in contrast to operationally defined theories of internal "mediating" states was not accidentally related to cognitive psychologists' rejection of behaviorist commitments to strong contiguity between cognitive and associative sensory-motor functions. Behaviorists justified their reductive explanations of human language learning and concept formation in terms of principles of conditioned learning derived from animal experimentation by maintaining that human cognitive functions differ only in degree of complexity from associative stimulus-response functions common to humans and animals. In contrast, cognitive theorists such as

Lashley, Chomsky, and Bruner claimed that language learning and concept formation could not be reductively explained as more elaborate forms of associative conditioned learning, but had to be explained in terms of the rule-governed processing of representational states. Behaviorists who treated cognitive states as nothing more than internal states that causally mediate between associated stimuli and responses were naturally inclined to operationally define them in terms of associated stimuli and responses. Cognitive psychologists who treated cognitive states as representational states processed in accord with rules were not even tempted to define them in terms of associated stimuli and responses.

COGNITION AND BEHAVIOR

There is another respect in which the cognitive revolution represented a fundamental discontinuity between earlier behaviorist and later cognitive forms of psychological theory. Cognitive psychologists from the 1950s onward did not merely aim to develop and evaluate cognitive explanations of behavior to replace explanations of behavior in terms of classical and operant conditioning: They also treated cognitive states and processes as legitimate objects of explanation in themselves. Although neobehaviorists treated references to cognitive states and processes as legitimate theoretical posits in the explanation of animal and human behavior, no behaviorist seems to have thought that cognitive states and processes were worthy of explanation or investigation in their own right. This was true even of Tolman, the most cognitive behaviorist, who gave the following examples of the types of behavior studied by behaviorist psychologists:

> A rat running a maze; a cat getting out of a puzzle-box; a man driving home to dinner; a child hiding from a stranger; a woman doing her washing or gossiping over the telephone; a pupil marking a mental-test sheet; a psychologist reciting a list of nonsense syllables; my friend and I telling one another our thoughts and feelings.
>
> —(1932, p. 8)

No reference was made to drawing an inference, solving an arithmetical problem, recalling the order of speakers at a wedding ceremony, or any other form of cognitive process.

The cognitive revolution in psychology involved the rejection of the behaviorist restriction of the subject matter of psychology to observable behavior. Cognitive psychologists since the 1950s have been as much concerned with the explanation of cognitive processing as they have been with the explanation of observable behavior. They have wanted to know whether images are processed phenomenally or sententially, whether semantic memory is accessed via the activation of hierarchically related conceptual nodes or feature lists, whether errors in inference are due to interference or the employment of distorting heuristics, and whether the prediction of the behavior of others is grounded in simulation of their cognitive processes or information-based theoretical modules. These sorts of questions were scarcely even raised, never mind addressed, within behaviorism.

Structuralism and Anthropomorphism

Their treatment of cognitive states and processes as legitimate objects of explanation does not mean that cognitive psychologists have neglected the critical evidential role of observable behavior in the empirical evaluation of theories of cognitive states and processes, and indeed cognitive psychologists often stress their employment of rigorous operational measures of cognitive constructs (Mandler, 1979). It also does not mean that the development of cognitive psychology marks a return to Titchener's structural psychology or Romanes's anthropomorphism, as various behaviorist critics of the cognitive revolution have charged (Amsel, 1989; Skinner, 1985, 1990).

Titchener's form of structural psychology was based upon two critical assumptions of the empiricist associationist tradition in psychology: that cognition is both imagistic and conscious. By contrast, the forms of cognitive psychology that developed from the 1950s onward did not equate cognition and imagery (Pavio, 1971) and did not presume that cognitive states and processes (including imagery) are conscious (Ericsson & Simon, 1980; Kosslyn, 1980). On the contrary, one of the heralded achievements of contemporary cognitive psychology has been the experimental demonstration that subjects have limited introspective access to their own cognitive states and processes:

> The accuracy of subject reports is so poor as to suggest that any introspective access that may exist is not sufficient to produce generally correct or reliable reports.
>
> —(Nisbett & Wilson, 1977, p. 233)

Neobehaviorists have complained that in attributing cognitive states and processes to animals, cognitive psychologists have reintroduced the methodological sins of "subjectivism" and "anthropomorphism" supposedly exorcised through the application of Morgan's canon:

> These animal cognitivists, who recognized a paradigm shift when they saw one, adopted the language and the models of human cognitive psychology and information processing. Consequently, instead of using animals as models for human function, they were now back to the kind of subjectivism and anthropomorphism the behaviorists had rejected. They were using animal behavior as a vehicle for their introspections, for understanding the mind, . . . and they were using humans as

> models for animal cognition. This use of humans as models for animals can happen *only in psychology*!
>
> —(Amsel, 1989, pp. 38–39)

Yet there is nothing intrinsically subjective or unscientific about using human cognition as a model for animal cognition. The only basis for calling such theoretical modeling subjective or unscientific is the supposed scientific illegitimacy of ascribing conscious or cognitive states to animals, according to the behaviorist interpretation of Morgan's canon. Yet although Morgan doubted that animals possess cognitive capacities such as means-end reasoning and abstract thought, he never doubted the scientific legitimacy of ascribing these capacities to them and maintained that there are good grounds for ascribing consciousness to animals. Behaviorists interpreted Morgan's canon as a prescription against the cognitive explanation of human and animal behavior only because they presumed that all human and animal behavior could be explained in terms of associative learning. Yet the work of Lashley, Chomsky, the Brelands, Garcia, Bruner, and Miller indicated that it could not.

Cognitive psychologists denied that cognitive processes could be explained in terms of classical and operant conditioning and thus denied the strong continuity between cognitive processes and reflexive associative processes affirmed by behaviorists (and by most late-19th-century physiologists and evolutionary theorists). Yet in ascribing cognitive processes to animals, cognitive psychologists also affirmed the strong continuity between human and animal psychology affirmed by behaviorists (and by most late-19th-century physiologists and evolutionary theorists). They did this without inconsistency because the denial of strong continuity between cognitive and reflexive associative processes does not mandate the denial of strong continuity between human and animal psychology and behavior. Although cognitive psychologists maintained (with Wundt and Morgan) that there are forms of cognition and patterns of behavior characteristic of humans that cannot be reduced to associative processes or conditioning, they also affirmed that many of these can be attributed to higher animals (Griffin, 1976; Premack & Premack, 1983; Roitblat, 1987). They maintained (contra Wundt and Morgan) that some animals had reached a stage of evolution sufficient for the emergence of such forms of cognition and patterns of behavior.

The Cognitive Tradition

Although the cognitive revolution did not represent a return to Titchener's structural psychology, it did represent a return to the "modern investigation of thinking" associated with the Würzburg school (Kulpe, 1912/1964). The Würzburg psychologists developed theories of cognitive processing that anticipated contemporary cognitive psychological theories: They postulated rule-governed operations on representational states without presupposing that such rules and representations were imagistic or conscious. Many pioneers of the cognitive revolution acknowledged the early work of the Würzburg psychologists. Newell, Shaw, and Simon's computer simulation of human problem solving was based upon the "thought-psychology" of Otto Selz, one of the members of the Würzburg school, and the work of his student Adriaan de Groot, who developed prototype "programs" for the analysis of problem solving by chess players (de Groot, 1946).

These early forms of cognitive psychology survived the demise of the Würzburg school in the 1920s. Thomas Vernon Moore (1877–1969), professor of psychology at the Catholic University of America, who visited the Würzburg psychologists in 1915, published *Cognitive Psychology* in 1938 (Knapp, 1985), and Woodworth's *Experimental Psychology* of that same year included a long chapter on "Thinking." A small group of American psychologists continued Wundt's experimental studies of attention throughout the early decades of the 20th century, until Broadbent and Miller returned them to center stage in the 1950s (Lovie, 1983, cited in Knapp, 1986). During the first half of the 20th century, psychologists continued to pursue cognitive psychological areas of research, such as problem solving, reasoning, concept formation, inductive inference, transfer of logical organization, and logical inference (Greenwood, 2001), and most of the experimental studies that supported these forms of research employed human subjects rather than rats and pigeons. A cognitive tradition developed from the 1920s onward that was continuous with contemporary cognitive psychology, albeit decidedly attenuated during the heyday of behaviorism.

Criticism and Connectionism

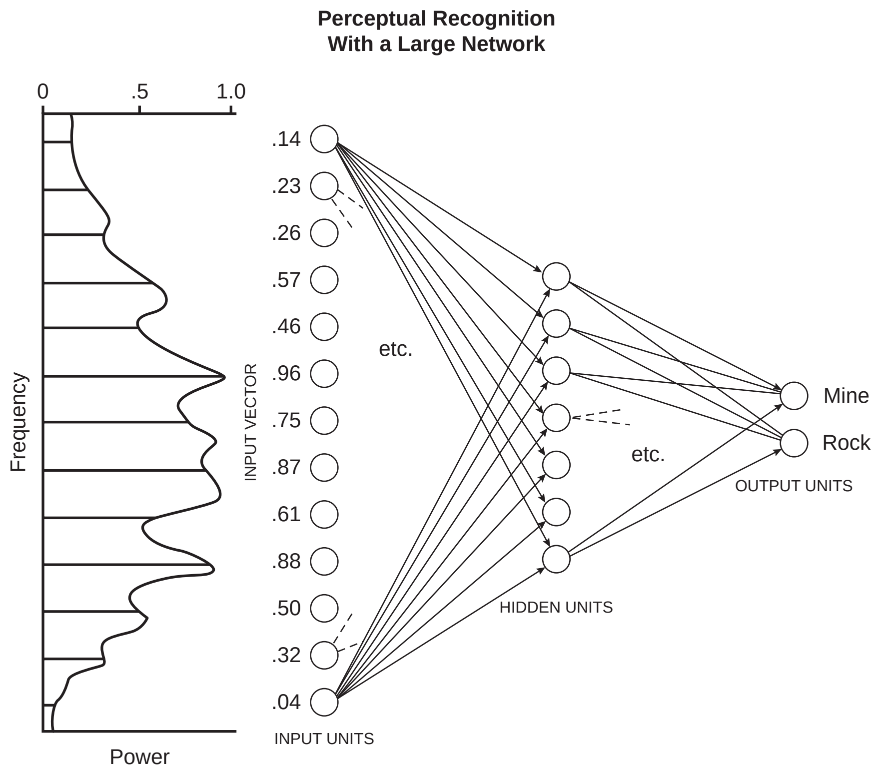
The cognitive revolution has not progressed entirely smoothly. Over and above behaviorist complaints about a return to subjectivism and anthropomorphism, a number of critics, including some of the early pioneers of the cognitive revolution, have raised concerns about the theoretical and empirical foundations of cognitive psychology. Some have complained about the fragmented nature of the discipline (Jenkins, 1981; Newell, 1973), noting that cognitive psychologists appear to study a wide variety of cognitive states and processes in the absence of any unified theory integrating research in perception, memory, and problem solving, for example. Yet some cognitive psychologists have developed integrative theories (Anderson, 1983), and it is possible that the lack of theoretical integration within cognitive psychology is itself a function of the limited integration of cognitive "modules" within cognitive architectures (Fodor, 1983). Others have complained about the artificiality of much experimental research in cognitive psychology, most notably Neisser, who called for a more "ecological" approach in his 1976 book *Cognition and Reality*. Yet although many critics have exploited his "second thoughts" to their rhetorical advantage, Neisser never questioned the basic theoretical or methodological integrity of cognitive psychology as a form of scientific psychology. He later affirmed that a wide range of current research in cognitive psychology is of "real significance" and that cognitive psychology and related disciplines "have made important and ecologically valid discoveries" (Neisser, 1997, p. 17).

A more interesting complaint was developed by John Anderson (1978, 1981), who claimed that behavioral data are insufficient to adjudicate between competing theories of representation, since a number of different theories of representation (in terms of propositions, images, or schema, for example) can accommodate the same behavioral data. Given this "fundamental indeterminacy" with respect to the empirical adjudication of competing theories, we are never in a position to infer the "psychological reality" of the relevant cognitive states and processes.

Yet the underdetermination of theories by empirical data is a fact of life in any science (Quine, 1960), and it is often possible to develop novel empirical predictions that enable scientists to empirically adjudicate between competing theories—as proved to be the case with respect to the conflict between the Ptolemaic and Copernican theories and the wave and particle theories of light. As Anderson himself acknowledged, competing theories of representation that are empirically underdetermined by behavioral predictions may be adjudicated by reference to neurophysiology (only one theory of cognitive processing may be consistent with the neural architecture of the brain) or by considerations of economy and efficiency. Given their rejection of introspective methods, cognitive psychologists cannot of course appeal to subject reports to establish the "psychological reality" of theories of representation. Yet in this respect they are no worse off than theoretical chemists or biologists, who likewise cannot appeal to the reports of molecules or cells to adjudicate between competing chemical and biological theories that predict the same range of empirical data.

It might turn out that the empirical underdetermination of competing cognitive theories is not a temporary function of the underdeveloped nature of theories of representation, but an enduring problem that cannot be resolved by appeal to novel prediction, neurophysiology, or considerations of economy and efficiency. Yet if this proved to be the case, it would be because a number of competing theories of representation have continued to furnish equally good explanations and predictions of a wide range of empirical behavior, are consistent with known neurophysiological data, and remain closely equivalent with respect to considerations of economy and efficiency. At this point cognitive psychology would have developed to a theoretical stage where it enjoys the luxury of a problem faced by advanced sciences such as subatomic physics.

Connectionism The most significant critique of cognitive psychology was internal and focused upon the reliance of theories of cognitive processing on theories of computation based upon the operation of von Neumann computers, which process information serially at high speeds. During the 1980s cognitive psychologists developed **connectionist** theories of cognitive processing, modeled upon **parallel distributed processing** (PDP) computer architectures, which encode information via the statistical distribution of connection "weights" among units in a nodal network. In a connectionist system, each unit is connected to other units and can send and receive excitatory and inhibitory signals. Connectionist systems process information when patterns of activation in designated "input" units spread through a system of internal units known as "hidden units," and patterns of activation stabilize in designated "output" units. In this fashion connectionist systems can distinguish rocks from mines, discriminate phonemes, and form the past tense of verbs. They excel in pattern recognition and learning, two areas that proved problematic for traditional cognitive systems based upon serial computation. The "Bible" of this new form of cognitive psychology was David Rumelhart and James



Parallel distributed processing: learning network for discriminating rocks from mines.

Churchland, P. (1988). Matter and consciousness. Cambridge, MA: MIT Press

McClelland's *Parallel Distributed Processing: Explorations in Microcognition,* which was published in 1986.

One of the perceived virtues of connectionist theories was that they were modeled upon the parallel processing of information by interconnected neurons in the brain:

> One reason for the appeal of PDP models is their obvious "physiological" flavor. They seem so much more closely tied to the physiology of the brain than other kinds of information-processing models. The brain consists of a large number of highly interconnected elements which apparently send very simple excitory and inhibitory messages to each other and update their excitations on the basis of these simple messages.
>
> —(McClelland, Rumelhart, & Hinton, 1986, p. 10)

Some have claimed that the development of connectionist theories has transformed cognitive psychology and marks the demise of traditional computational theories of cognition, since connectionist theories of processing do not presuppose the cognitive representation of rules, a distinctive feature of cognitive theories from Chomsky to Neisser. For example, commenting on the performance of a connectionist network that simulated the acquisition of English past tense, Rumelhart and McClelland claimed that

> We have, we believe, provided a distinct alternative to the view that children learn the rules of English past-tense formulation in any explicit sense. We have shown that a reasonable account of the acquisition of past tense can be provided without recourse to the notion of a rule as anything more than a *description* of the language.
>
> —(Rumelhart & McClelland, 1986, p. 267)

However, connectionist theories do not deny the representation of rules, but only particular theories about how they are encoded and learned. They claim that rules are represented within the distribution of connection "weights" between units and that networks learn via the modular adjustment of the strength of connections between units, in contrast to traditional computational cognitive theories based upon von Neumann architectures, in which a central processing unit controls the processing of information in accord with a set of rules stored in the computer program.

Consequently, it may be argued that connectionism is better represented as a creative development or transformation of traditional cognitive theories than as a displacement of them. This seems to have been the original view of the pioneers of connectionist theory:

> Though the appeal of PDP models is definitely enhanced by their physiological plausibility and neural inspiration, these are not the primary basis for their appeal to us. We are, after all, cognitive scientists, and PDP models appeal to us for psychological and

> computational reasons. They hold out the hope of offering computationally sufficient and psychologically accurate mechanistic accounts of the phenomena of human cognition which have eluded successful explication in conventional computational formalisms; and they have radically altered the way we think about the time-course of processing, the nature of representation, and the mechanisms of learning.
>
> —(McClelland, Rumelhart, & Hinton, 1986, p. 11)

Defenders of traditional cognitive psychology have maintained that connectionism poses no threat to traditional cognitive theories, either because they doubt the ability of connectionist systems to "scale up" to cover more complex and integrated cognitive processes (Dennett, 1991) or because they treat connectionist theories as merely accounts of how the forms of information processing described by traditional cognitive theories are implemented in the human brain (Fodor & Pylyshyn, 1988). While some have suggested that developments in connectionism and (more recently) "dynamical-systems" theory (Port & Gelder, 1995; Serra & Zanarini, 1990) mark a return to earlier forms of associationist psychology and behaviorism (Haselager, 1997), contemporary connectionist theories based upon computational learning algorithms appear to bear only a very tenuous relation to earlier traditions based upon the correlation of imagistic ideas and conditioned stimulus response connections.

THE SECOND CENTURY

One thing is for sure. The cognitive revolution is an ongoing revolution, and theories of cognitive processing continue to be developed in creative and fertile ways. Moreover, whatever one concludes about the significance of recent developments in connectionist theory, it is clear that contemporary scientific psychology has reached a stage where it does not consider the postulation of cognitive theories (about unconscious and unobservable cognitive states) to be unscientific and no longer maintains that the content of scientific psychological theories should be restricted to the content of operational definitions. Consequently, at the beginning of the 21st century, with psychology having celebrated the first centennial of scientific psychology in 1987 and of the APA in 1992, we may close this conceptual history of psychology on a generally positive note.

Without presuming a presentist perspective on the development of psychological theory and methodology, it can be said that contemporary psychology embraces roughly the same conception of scientific theory as the mature physical sciences, even though it continues to romanticize about its ability to reproduce particular (and arguably inappropriate) features of Newtonian science, such as universal explanation (Kimble, 1995; Shepard, 1987, 1995) and unified theory (Spence, 1987; Staats, 1983). Perhaps in the 21st century psychology will finally free itself of this conceptual baggage from its early protoscientific history.

It may be forced to, since the fate of cognitive psychology is no longer exclusively in the hands of psychologists. Cognitive psychology now forms part of the broader interdisciplinary matrix that is cognitive science, which will undoubtedly be influenced by developments in the other disciplines (such as computer science, neuroscience, linguistics, mathematics, and philosophy) that make up cognitive science. What is true of the fate of cognitive psychology is also true of the fate of institutional psychology, since the APA no longer exclusively determines the future of the discipline.

The APA grew at a phenomenal rate in the postwar period: from over 9,000 in 1950 to over 19,000 in 1960; from over 30,000 in 1970 to over 50,000 in 1980; and from over 70,000 in 1990 to over 83,000 in 2000 (APA, 2001). The charter members of the APA met in G. Stanley Hall's office at Clark University in 1892. Today conference attendees number between 10,000 and 20,000 and occupy a slew of hotels in major cities. APA revenues have grown from just over $60 in 1892 to tens of millions of dollars in the 21st century.

Most of the growth in the postwar period came from professional rather than academic psychology, and the balance between the academic and professional psychologists within the APA began to shift in favor of the professionals (clinical, counseling, industrial, educational, and other applied psychologists). Before the war about two thirds of the APA membership were academics; by the 1980s only about a third were. Throughout the 1960s and 1970s academic psychologists pressed for changes in the structure of the APA that would have given them greater power within an organization increasingly dominated by professionals. When the APA rejected a restructuring plan in 1988, the academics split to form the American Psychological Society (APS). From an initial charter membership of 1,500, the APS grew to 5,000 by 1989, and to 15,000 by 1995 (Brewer, 1994). The membership of the APS, unlike the membership of the APA, includes many biologists, computer scientists, philosophers, linguists, and mathematicians, who are not academically qualified psychologists (who do not have a degree in psychology). Given that the qualified memberships only partially overlap, it is unlikely that the two associations will reunify in the near future.

DISCUSSION QUESTIONS

1. Do you think that some machines are intrinsically purposive? Do we ascribe purpose to machines because they employ feedback or because purposive humans design them? If a machine malfunctioned in a systematic way, would we still call its behavior purposive?
2. Newell, Shaw, and Simon suggested that the adequacy of the computer simulation of human problem solving depends upon the ability to mimic the

cognitive strategies of human problem solvers, as revealed by "think-aloud" subject protocols. Is this a legitimate criterion of adequacy for the computer simulation of cognition? What problems might you anticipate with such an approach?

3. Contemporary cognitive psychology appears to be committed to strong continuity between human and animal psychology but not to strong continuity between higher cognitive and lower reflexive associative processes. Is this position consistent?
4. Would a computer programmed to operate on principles of classical and operant conditioning be able to understand language or engage in purposive behavior?
5. Do you think it possible to reconstruct the cognitive revolution as a Kuhnian "paradigm shift"?

GLOSSARY

artificial intelligence The science of intelligent machines.

bit A binary unit, the elemental unit of information theory, conceived of as the amount of information required to determine between two equiprobable alternatives.

cognitive science The name for the interdisciplinary matrix of cognitive psychology, artificial intelligence, and linguistics (and associated disciplines such as neurophysiology, logic, mathematics, and philosophy) that evolved as a product of the cognitive revolution in psychology.

connectionism Set of cognitive theories developed in the 1980s modeled upon the parallel processing of information in the brain rather than the serial processing of von Neumann computers.

cybernetics The science of control and communication in animals and machines.

digital computer A computer that performs operations on binary units of information.

feedback Signals from a goal that modify goal-directed behavior.

General Problem Solver Computer designed by Allen Newell, J. C. Shaw, and Herbert Simon, capable of cognitive tasks such as playing chess and problem solving.

hardware The physical systems that instantiate the rules or instructions encoded in computer programs.

Logic Theorist Computer program designed by Allen Newell and Herbert Simon, capable of proving theorems in logic.

new logic Form of truth-functional logic developed in the late 19th and 20th centuries, whose sentential operators (such as "if . . . then") are definable in terms of primitive operators such as "and" and "not."

noise Random disturbance superimposed upon a signal, such as electrical noise caused by heat in electrical circuits.

parallel distributed processing Computer architecture in which information is encoded via the statistical distribution of connection "weights" among units in a nodal network.

program A set of rules or instructions stored in the memory of a computer.

servomechanism Term used to describe intrinsically purposive machines such as torpedoes, whose behavior is regulated by feedback.

software The set of rules or instructions encoded by a computer program.

strong AI View that we can ascribe cognitive states to computers in exactly the same sense that we ascribe them to humans.

Turing machine A machine capable of performing elementary operations on symbols in accord with a set of instructions.

Turing test A hypothetical test suggested by Alan Turing as a practical means of deciding whether intelligence should be ascribed to a machine. Turing claimed that we should call a machine intelligent if we could not discriminate the responses of a human communicator from a machine simulating or "imitating" the responses of a human communicator.

universal Turing machine A general-purpose machine capable of performing operations specified by a variety of different Turing machines (defined by their individual sets of instructions).

von Neumann machine Computer that employs a central control unit to read and execute programmed instructions sequentially.

wetware Term used to describe biological systems that can instantiate the rules or instructions encoded in computer programs.

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CHAPTER 13

Abnormal and Clinical Psychology

HISTORIES OF ABNORMAL AND CLINICAL PSYCHOLOGY TEND TO BE decidedly presentist and generally represent the history of psychological theory and therapy as a progression from superstitious theories of spirit possession and brutal persecution to contemporary scientific theories and humane treatments (Sedgewick, 1982). Often this is based upon little more than condescending assumptions about earlier peoples. For example, when Neolithic skulls were discovered with holes in them, the French neurophysiologist Paul Broca opined that they must have been made "in order to liberate evil spirits" (Ackerknecht, 1982, pp. 8–9), a representation still repeated in contemporary histories of psychology. Yet it is just as likely that such holes were the product of early forms of trepanning, the removal of part of the skull to reduce swelling of the brain caused by injury through war or hunting.

As noted earlier, the ancient and medieval peoples were rather more sophisticated and humane, at least relative to their times, than they are usually given credit for. They recognized depression, mania, and hysteria and attributed most psychological disorders to neural causes. Their psychological treatments, which were usually based upon holistic principles derived from the Hippocratic school and Galen, included fresh air, relaxation, dieting, and music, as well as bloodletting and purgation.

From early Roman times, laws governed the treatment of the psychologically disturbed and provided for family or community guardianship of persons designated as "insane" or "mad" (Neaman, 1975). These laws recognized that such persons were not legally responsible for their actions, because of their diminished or defective powers of reasoning (Maher & Maher, 1985a; Neugebauer, 1978). Institutions devoted to the treatment of the psychologically disturbed were set up in medieval cities such as Baghdad, Valencia, and London, and the common law of many medieval European states included protections for them (Schoeneman, 1977).

Those deemed insane were generally cared for by their families or legally appointed guardians. They were occasionally beaten—the common law of England allowed people to beat their mad relatives (Allderidge, 1979)—and those deemed dangerous were restrained or imprisoned. Although no doubt many were persecuted